TorontoXP Sports Activity Finder

Product Requirements Document

Document Type: Retrospective PRD / Product Specification
Product: TorontoXP Sports Activity Finder
Status: Completed
Platform: Progressive Web Application (PWA)
Version: 1.3
Last Updated: 03-Oct-2026

1. Purpose of the document

This document serves as a retrospective specification documenting the current functional baseline of the TorontoXP Sports Activity Finder (v1.3) as of October 2026.

2. Overview & Problem Statement

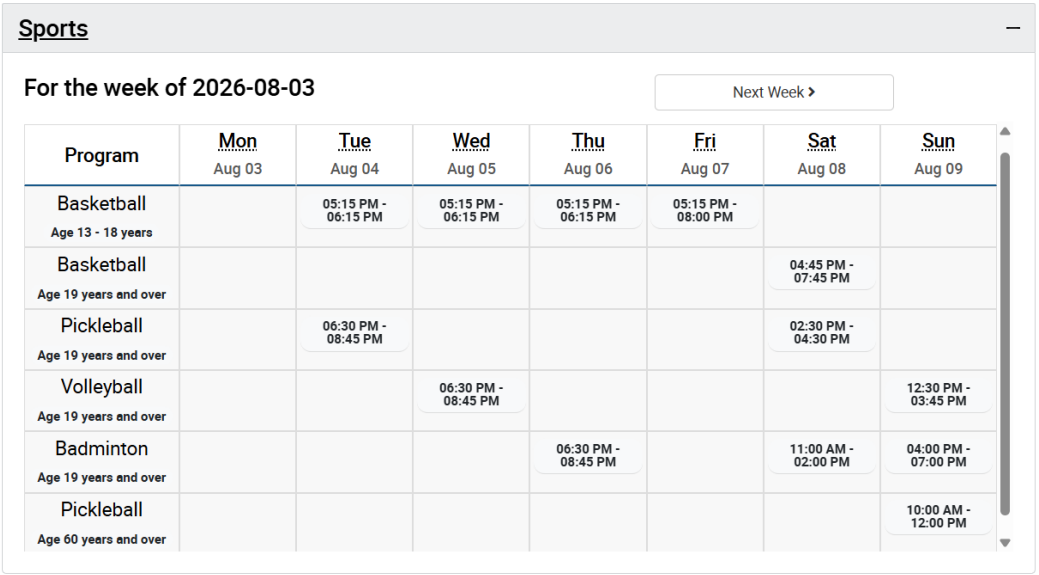
1.1 Context:

The official City of Toronto website provides individual pages for each community centre, with schedules listing the activities and programs offered at that specific location.

1.2 Problem Statement

The current recreation schedule experience is optimized for users who already know which community centre they want to visit. However, there is no reverse-search capability that allows users to discover community centres based on the sport or activity they are interested in.

For example, a user looking to play basketball or badminton must manually visit and review individual community centre schedules to determine which locations offer the desired activity. This creates unnecessary effort and makes it difficult for users to discover suitable locations across the city.



Sports

For the week of 2026-08-03

Next Week ›

Program	Mon Aug 03	Tue Aug 04	Wed Aug 05	Thu Aug 06	Fri Aug 07	Sat Aug 08	Sun Aug 09
Basketball Age 13 - 18 years		05:15 PM - 06:15 PM	05:15 PM - 06:15 PM	05:15 PM - 06:15 PM	05:15 PM - 08:00 PM		
Basketball Age 19 years and over						04:45 PM - 07:45 PM	
Pickleball Age 19 years and over		06:30 PM - 08:45 PM				02:30 PM - 04:30 PM	
Volleyball Age 19 years and over			06:30 PM - 08:45 PM				12:30 PM - 03:45 PM
Badminton Age 19 years and over				06:30 PM - 08:45 PM		11:00 AM - 02:00 PM	04:00 PM - 07:00 PM
Pickleball Age 60 years and over							10:00 AM - 12:00 PM

Figure 1 Typical sports schedule from a community centre website

1.3 Key Pain Points

- Sport discovery is **centre-first rather than sport-first**.
- Users must inspect multiple community-centre schedules individually.

- There is no centralized view of sessions for a specific sport.
- Sport and activity names are not always consistent across locations.
- Comparing sessions across locations, days, age groups, and costs is difficult.
- Important actions such as finding directions or contacting a centre require additional navigation.

3. Product Opportunity

The core opportunity was to create a **reverse-search experience**:

*Instead of asking "**What sports are available at this community centre**?", allow users to ask "**Where can I play this sport**?"*

2.1 Information Architecture Analogy — Transposing a Matrix

Imagine the existing recreation data as a matrix where:

- **Rows = Community Centres**
- **Columns = Sports / Activities**
- **Cells = Sessions or availability**

The City of Toronto website exposes this information primarily row-by-row: users select a community centre and then explore its activities.

TorontoXP conceptually transposes this discovery model, allowing users to start with a sport and discover the community centres and sessions associated with it.

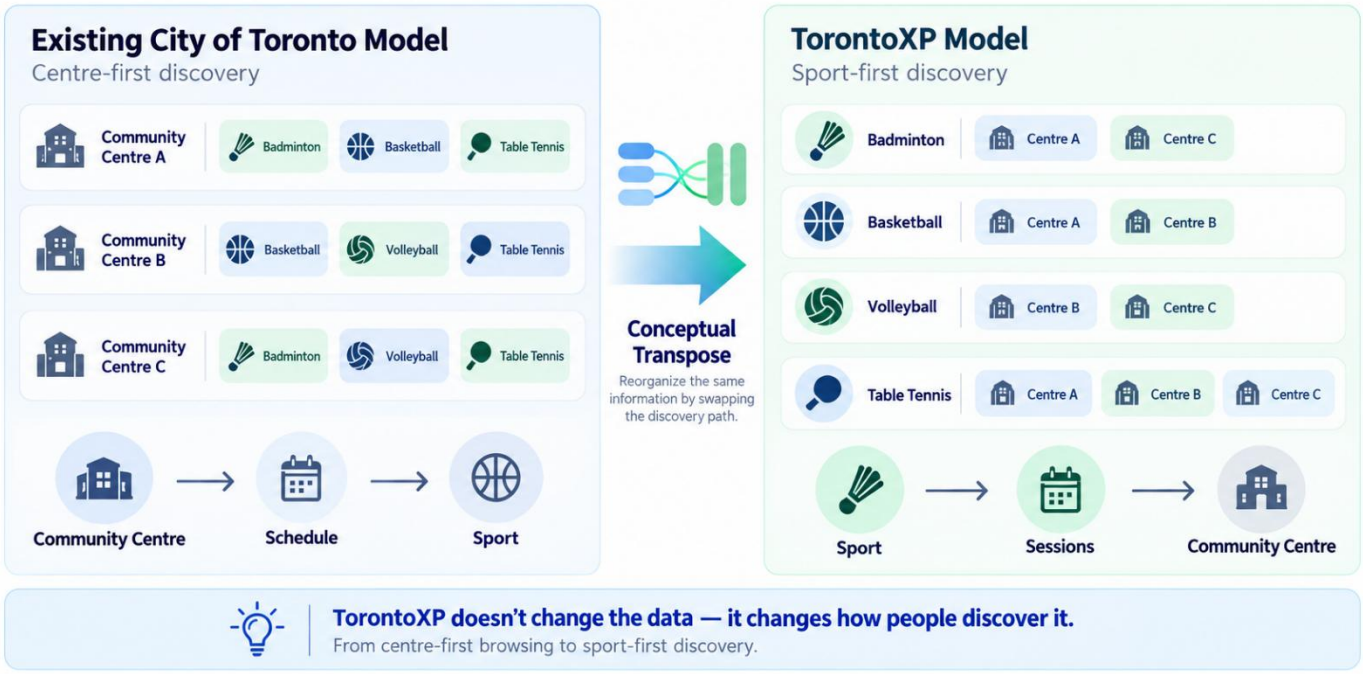


Figure 2 Transposing Matrix Analogy Representation

4. Target Users

- **Adult Sports Enthusiasts**: Casual and competitive players looking for drop-in sessions for specific sports.
- **Parents**: Searching for affordable sports programs and open gym times for their children (kids, teens).
- **Seniors**: Looking for accessible activities like Pickleball, Bocce, or Carpet Bowling.

- **Budget-conscious residents**: People seeking free or low-cost recreational activities.

5. Product Goals, Scope & Non-Goals

5.1 Primary Goal

- Enable Toronto residents to discover community-centre sports by **sport first**, rather than requiring them to search by location.

5.2 In Scope

- The current product covers **sports and sport-related recreational activities** offered through Toronto community centres.
  Examples: Basketball, Badminton, Pickleball, Squash, Volleyball, etc.

5.3 Future Scope

- The underlying approach could eventually be extended to other recreation categories, including:
  - Swimming
  - Fitness
  - Skating
  - Arts, Hobbies & Interests

  These categories are intentionally excluded from the current implementation to maintain a focused product scope and establish the sport-discovery experience first.

5.4 Non-Goals

- The product is intended to be a **discovery and navigation layer**, rather than a replacement for the City's official recreation system.

  The following are outside the current scope:
  - Processing registrations or payments.
  - Replacing the City's official booking system.
  - Managing user accounts.
  - Guaranteeing real-time session availability or capacity.
  - Managing community-centre schedules directly.
  - Providing customer support for City programs.

  The official City website remains the authoritative source for registration and program details.

6. Core User Stories

6.1 Sport-First Discovery

- As a Toronto resident, I want to start my search with a sport so that I can discover where I can play it.

6.2 Schedule Discovery

- As a user, I want to see all available sessions for a sport so that I can find a suitable day and time.

6.3 Filtering

- As a user, I want to filter sessions by day, age group, cost, and program type so that I can quickly identify relevant options.

6.4 Sorting

- As a user, I want to sort the community centres based on their distance from my location and also sort the soonest upcoming activity.

6.4 Taking Action

- As a user, I want to get directions, call the community centre, or visit the official program page so that I can act on the information I found.

7. Key Product Requirements

- **Sport-Centric Discovery**: Users can browse a grid of available sports and select one to see all active sessions across the city.
- **Smart Filtering**: Users can drill down into schedules using multiple filters:
  - Day of the Week
  - Age Group (e.g., 19+, 6-12)
  - Cost (Free vs. Paid)
  - Program Options / Tags (e.g., Women only, Family, Youth Zone, Caregiver)
  - Specific Community Centre(s)
- **Direct Actions**: From the schedule table, users can:
  - Open the community centre location directly in Google Maps.
  - Call the community centre directly via a click-to-call button.
  - Visit the community centre's official webpage for registration details.
- **Automated Data Pipeline**: A backend script scrapes the city's website weekly (every Monday), using AI (Gemini 2.5 Flash or other free model) to normalize activity names (e.g., mapping "Parasport: Wheelchair Basketball" to "Basketball" with a tag) and seeds a DynamoDB database.

8. Product Analytics Strategy

The product uses privacy-conscious analytics tools, including:

- TelemetryDeck
- GoatCounter
- Simple Analytics

The analytics strategy focuses on **aggregated product behaviour** to inform future product decisions rather than collecting unnecessary personal information.

Key questions the telemetry might answer include:

1. What sports are users most interested in?
2. Which filters are most valuable?
3. Which sessions generate the most outbound actions?
4. Are users using the product primarily for discovery or for direct navigation to City resources?

9. Technology Stack

- **Frontend**: React 19, Vite 8, React Router, deployed on GitHub Pages.
- **Backend (Serverless)**: AWS API Gateway, AWS Lambda (Python 3.12).
- **Database**: AWS DynamoDB.
- **Data Pipeline**: Python web scraper + Gemini 2.5 Flash API for data normalization and tagging.
- **Infrastructure**: AWS SAM for infrastructure as code.

10. Solution

TorontoXP Sports Activity Finder is PWA that lets users discover drop-in sports programs at City of Toronto community centres. It aggregates schedule data from all community centres, normalizes it, and presents it in a ***"sport-first"*** interface, allowing users to effortlessly find where they can play their favourite sports across Toronto.

Below are a few screenshots of the web app in desktop & mobiles.

